Roll No. | | | | | | | |

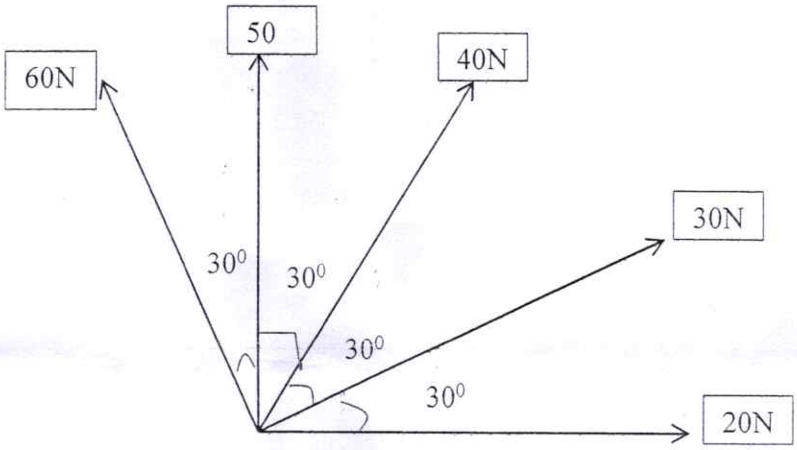
Mid Semester Examination June 2025

Name of the Course: B. Tech Civil Engg Semester: III
Name of the Paper: Mechanics of Solids Course Code: TCE 303
Time: 90 minutes Maximum Marks: 50

Note: -
(i) **Answer all the questions by choosing any one of the sub questions.**
(ii) **Each question carries 10 marks**

Q1 **(10marks)**

(a) The forces 20N, 30N, 40N, 50N and 60N are acting at one of the angular point, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force. **CO1**



or

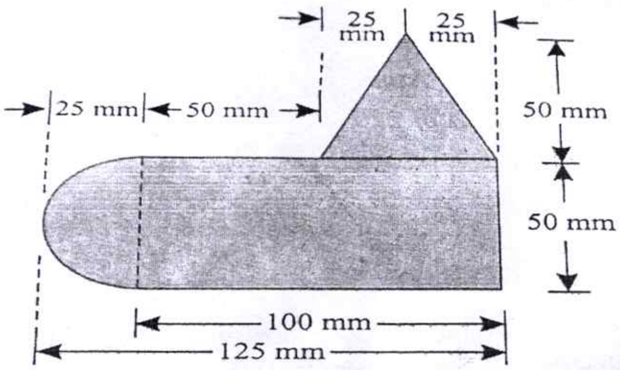
(b) Find the Young's Modulus of a brass rod of diameter 25 mm and of length 250 mm which is subjected to a tensile load of 50 kN when the extension of the rod is equal to 0.3 mm.

Q2 **(10 marks)**

(a) A steel rod of 20 mm diameter passes centrally through a copper tube of 50 mm external diameter and 40 mm internal diameter. The tube is close at each end by rigid plates of negligible thickness. The nuts are tightened lightly home on the projecting parts of the rod. If the temperature of the assembly is raised by 50 ^{0}C, calculate the stresses developed in copper and steel. Take E for steel and copper as 200 GN/m^2 and 100 GN/m^2 and α for steel and copper as 12×10^{-6} per ^{0}C and 18×10^{-6} per ^{0}C. **CO1**

or

(b) A uniform lamina shown in figure consists of a rectangle, a circle and a triangle. Determine the centre of gravity of the lamina. All dimensions are in mm.



Q3 (10 marks)

(a) Find the expression for the total extension of bar due to its self-weight, when the bar is fixed at its upper end and hanging freely at the lower end. CO1, CO2

or

(b) Determine the Poisson's ratio and bulk modulus of a material, for which young's modulus is 1.2×10^5 N/mm^2 and modulus of rigidity is 4.8×10^4 N/mm^2.

Q4 (10 marks)

(a) Define the terms: Elasticity, Hardness, Young's modulus, Modulus of rigidity and Hooks law. CO2

or

(b) Draw the stress and strain graph of mild steel, define and mention each terms.

Q5 (10 marks)

(a) Derive the equation of extension of uniformly tapering circular rod with diameter D1 and D2. Assume suitable dimensions and parameters. CO2

or

(b) Describe the equation for the total elongation of a uniformly tapering rectangular bar when subjected to axial load P. Assume suitable dimensions and parameters.